

Name:	
Class:	



Year 12
Assessment task 2
Term 1 2018

MATHEMATICS EXTENSION 1

Time Allowed - 120 minutes

(plus 5 minutes Reading Time)

General Instruction:

- All questions may be attempted
- All questions are equal value
- Reference Sheet will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or badly arranged work

Questions 1-5 are to be completed on the multiple choice sheet, then Questions 6-12 to be completed on separate sheets of paper and returned in separate bundles. Questions must show your Candidate Number (in the top right hand corner).

SECTION I
MULTIPLE CHOICE QUESTIONS

Attempt Questions 1 – 5 (1 mark each).

Allow approximately 8 minutes for this section

1. The value of the expression

$$36^{\log_6 5} + 10^{1-\log_{10} 2} - 3^{\log_9 36}$$

is:

- A) $\frac{74}{3}$;
- B) 24;
- C) $\frac{73}{4}$;
- D) 6.

2. Which of the following is an expression for $\int \cos^2 2x dx$?

- A) $x - \frac{1}{4} \sin 4x + c$;
- B) $x + \frac{1}{4} \sin 4x + c$;
- C) $\frac{x}{2} - \frac{1}{8} \sin 4x + c$;
- D) $\frac{x}{2} + \frac{1}{8} \sin 4x + c$;

3. The graph of the function $f(x) = e^x \sin x$, $0 \leq x \leq \pi$, has a maximum gradient of:

- A) 1;
- B) $\frac{\pi}{2}$;
- C) e^π ;
- D) $e^{\frac{\pi}{2}}$.

4. The angle between tangents of the graph $y = 2x^3 + x$ at the points $(-1, -3)$ and $(1, 3)$ is:

- A) $\frac{\pi}{3}$;
- B) $\tan^{-1}(6)$;
- C) 0;
- D) $\frac{\pi}{2}$

5. Four functions $f_1(x) = 2 \log_2 x$, $f_2(x) = \log_2 x^2$, $f_3(x) = 2 \log_2 |x|$ and $f_4(x) = \frac{2}{\log_x 2}$ are given, find the true statement:

- A) $f_1 \neq f_2 = f_3 \neq f_4 \neq f_1$;
- B) $f_2 \neq f_1 = f_4 \neq f_3 \neq f_2$;
- C) $f_3 \neq f_1 = f_2 \neq f_4 \neq f_3$;
- D) all four functions are equal.

END OF SECTION I

SECTION II
EXTENDED RESPONSE (77 marks)

77 marks**Attempt Questions 6 – 12****Allow approximately 1 hour and 52 minutes for this section****MARKS:****QUESTION 6. (11 marks)****COMMENCE A NEW PAGE**

(a) Find

$$\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{2x} \right)$$

1(b) Using the substitution $u = 1 + e^x$, find the value of $\int \frac{e^{3x}}{1 + e^x} dx$.**3**(c) Use mathematical induction to prove that $3^{2n} - 1$ is divisible by 8, for all integers $n \geq 1$.**4**

(d) A ladder 3 metres long has its top end resting against a vertical wall and its lower end on horizontal ground. The top end of the ladder is slipping down at a constant rate of 0.1 metres per second. Find the rate at which the lower end is moving away from the wall when the lower end is 1 metre from the wall.

3**QUESTION 7. (11 marks)****COMMENCE A NEW PAGE**

(a) Consider the function

$$f(x) = \cos x + \sqrt{3} \sin x, \quad 0 \leq x \leq 2\pi$$

Given that $f(x)$ can be expressed in the form $r \cos(x - a)$, where $r > 0$ and $0 < a < \frac{\pi}{2}$:(i) Show that the values of $r = 2$ and $a = \frac{\pi}{3}$.**2**

(ii) Find the range of the function.

1(iii) Find the coordinates of the y -axis intercept.**1****QUESTION 7. (continues on next page)**

QUESTION 7. (continues from previous page)**MARKS:**

- (iv) Find the coordinates of the x -axis intercept.

2

- (v) Graph $f(x) = \cos x + \sqrt{3}\sin x$.

2

- (vi) Find the volume of the solid formed when the region bounded by the graph of $y = f(x)$, the x -axis and y -axis is rotated about the x -axis.

3**QUESTION 8. (11 marks)****COMMENCE A NEW PAGE**

- (a) Let $f(x) = x \log_e(x - 1)$. What is the domain and range of $f(x)$?

2

- (b) (i) Find the value of a such that

$$\frac{1}{x+1} - \frac{1}{x+4} = \frac{a}{(x+1)(x+4)}.$$

1

- (ii) Hence or otherwise, find the value of

$$\int_0^1 \frac{dx}{x^2 + 5x + 4}$$

2

- (c) Use substitution $x = 2 \sin \theta$ to evaluate $\int_0^1 \sqrt{4 - x^2} dx$.

3

- (d) (i) Find the derivative of $f(x) = \tan(x^2)$.

1

- (ii) Hence or otherwise evaluate $\int_{-a}^a x \sec^2(x^2) dx$.

2

MARKS:**QUESTION 9. (11 marks)****COMMENCE A NEW PAGE**

- (a) In the arithmetic sequence $T_4 = 5$. Find the value of the common difference, d , when the sum of the squares of the first three terms in the sequence is a minimum.

3

- (b) Prove by mathematical induction that for all positive integer values of n ,

$$\frac{1}{3} \times \frac{1}{1} + \frac{1}{5} \times \frac{1}{3} + \frac{1}{7} \times \frac{1}{5} + \dots + \frac{1}{(2n+1)} \times \frac{1}{(2n-1)} = \frac{n}{2n+1}.$$

4

- (c) Sketch the graph of

$$y = \frac{24}{x^2 + 2x - 24}.$$

State the equations of any asymptotes. Find the coordinates of any axis intercepts and any turning points.

4**QUESTION 10. (11 marks)****COMMENCE A NEW PAGE**

- (a) (i) Show that the equation of the tangent to the parabola $x^2 = 4Ay$ at any point $P(2Ap, Ap^2)$ is given by $px - y - Ap^2 = 0$.

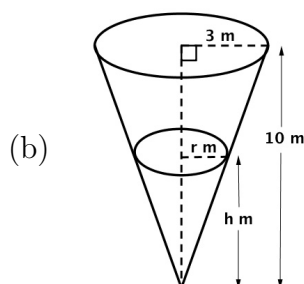
2

- (ii) S is the focus of the parabola and T is the point of intersection of the tangent and the y -axis. Prove that $SP = ST$.

3

- (iii) Hence show that $\angle SPT$ is equal to the acute angle between the tangent and the line through P parallel to the axis of the parabola.

2**QUESTION 10. (continues on next page)**

QUESTION 10. (continues from previous page)**MARKS:**

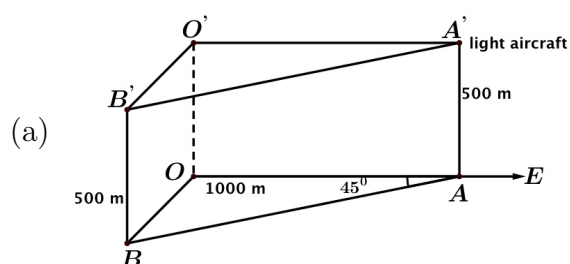
The diagram shows a conical wheat flu. The flu is being filled with wheat at the rate of 2 m^3 per minute. The height of wheat at time t minutes is h metres and the radius of the wheat's top surface is r metres.

(i) Show that $r = \frac{3h}{10}$.

1

(ii) Find the rate at which the height is increasing when the height of the wheat is 8 m.
(Volume of cone = $\frac{1}{3}\pi r^2 h$).

3

QUESTION 11. (11 marks)**COMMENCE A NEW PAGE**

A light aircraft flying at a height of 500 m above the ground is sighted by an observer stationed at a point, O , on the ground, 1 km due west of the plane. The aircraft is flying south-west from A' (along $A'B'$) at 300 km/h.

(i) How far will it travel in 1 minute?

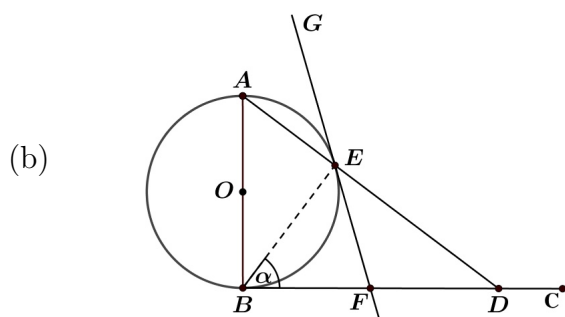
1

(ii) Find its bearing from OO' at this time.

3

(iii) What will be its angle of elevation from O at this time?

2



In the diagram, AB is a diameter of the circle, centre O , and BC is tangential to the circle at B . The line AED intersects the circle at E and BC at D . The tangent to the circle at E intersects BC at F . Let $\angle EBF = \alpha$.

(i) Copy the diagram into your Writing Booklet labelling all appropriate information.

1

(ii) Prove that $\angle FED = \frac{\pi}{2} - \alpha$.

2

(iii) Prove that $BF = FD$

2

MARKS:**QUESTION 12. (11 marks)****COMMENCE A NEW PAGE**

(a) The cylinder, with the radius x and height H , is inscribed in a cone of height h .

(i) If θ is semi-vertical angle of the cone, show that the volume of cylinder is given by:

$$V = \pi(x^2h - x^3 \cot \theta)$$

2

(ii) Show that for the maximum volume of the inscribed cylinder the height of the cylinder is the third of the height of the cone, $H = \frac{1}{3}h$.

3

(b) (i) Write down an expression for $\cos(a + b)$ and hence prove that $\cos(2q) = 1 - 2\sin^2 q$.

1

(ii) Prove the identity

$$\frac{\cos y - \cos(y + 2q)}{2 \sin q} = \sin(y + q).$$

2

(iii) Use mathematical induction and the result of part (ii) to prove the identity:

$$\sin q + \sin 3q + \sin 5q + \dots + \sin(2n - 1)q = \frac{1 - \cos 2nq}{2 \sin q}.$$

3**END OF THE TEST**



Student Number 222

Multiple Choice

- (1) B
- (2) D
- (3) D
- (4) C
- (5) A

MATHEMATICS: Question.....

Suggested Solutions	Marks Awarded	Marker's Comments
<u>Question 6</u> a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$ $= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{2}$ $= \frac{3}{2} \times \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$ $= \frac{3}{2} \times 1$ $= \frac{3}{2}$ b) $\int \frac{e^{3x}}{1+e^x} dx$ let $u = 1+e^x$ $\frac{du}{dx} = e^x$ $dx = \frac{du}{e^x}$ $= \frac{du}{u-1}$ $\int \frac{e^{3x}}{1+e^x} dx$ $= \int \frac{(u-1)^3}{u} \times \frac{du}{u-1}$ $= \int \frac{(u-1)^2}{u} du$ $= \int \frac{u^2 - 2u + 1}{u} du$ $= \int u - 2 + \frac{1}{u} du$ $= \frac{u^2}{2} - 2u + \ln u $	1 mark for differentiating correctly and making correct substitution. Note: students lost a mark for mixing variables in their integral	1 mark for answer although students should show working out very well done 1 mark for integrating correctly

MATHEMATICS Extension 1 : Question.....

Suggested Solutions

Marks Awarded

Marker's Comments

$$= \frac{(1+e^x)^2}{2} - 2(1+e^x) + \ln(1+e^x) + C$$

c) when $n=1$,

$$3^{2 \times 1} - 1$$

$$= 9 - 1$$

$$= 8$$

$$= 8 \times 1$$

$\therefore$ divisible by 8

$\therefore$ statement is true for $n=1$

Assume true for $n=k$ where $k \in \mathbb{Z}$

$$3^{2k} - 1 = 8M \text{ where } M \in \mathbb{Z}$$

Prove for $n=k+1$

$$3^{2(k+1)} - 1$$

$$= 3^{2k+2} - 1$$

$$= 9 \times 3^{2k} - 1$$

$$= 9 \times (8M+1) - 1 \text{ (by assumption)}$$

$$= 72M + 9 - 1$$

$$= 72M + 8$$

$$= 8(9M+1)$$

$$= 8P \text{ where } P = 9M+1$$

Since $M \in \mathbb{Z}$, $9M+1 \in \mathbb{Z}$

$\therefore$ divisible by 8 when $n=k+1$

$\therefore$ statement is true by the principle of mathematical induction

1 mark for remembering to resubstitute u .

1 mark for showing true for $n=1$

1 mark for correct assumption

1 mark for stating $k \in \mathbb{Z}$

1 mark for correct steps proving true for $n=k+1$

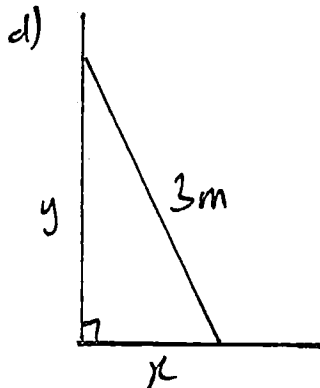
MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks Awarded

Marker's Comments

d)



$$\frac{dy}{dt} = -0.1 \text{ m/s}$$

$$\frac{dx}{dt} = ?$$

$$\frac{dx}{dt} = \frac{dx}{dy} \times \frac{dy}{dt}$$

$$x^2 + y^2 = 9 \quad (\text{Pythagoras})$$

$$y^2 = 9 - x^2$$

$$y = \sqrt{9 - x^2} \quad (y > 0)$$

$$\frac{dy}{dx} = \frac{1}{2} (9 - x^2)^{-\frac{1}{2}} \times (-2x)$$

$$= -\frac{x}{\sqrt{9 - x^2}}$$

$$\frac{dx}{dt} = -\frac{\sqrt{9 - x^2}}{x} \times (-0.1)$$

when $x = 1$,

$$\frac{dx}{dt} = \frac{\sqrt{9 - 1^2}}{1} \times 0.1$$

$$= \frac{\sqrt{8}}{10}$$

$$= \frac{2\sqrt{2}}{10}$$

$$= \frac{\sqrt{2}}{5} \text{ m/s}$$

$\therefore$ the lower end is moving away at $\frac{\sqrt{2}}{5} \text{ m/s}$.

1 mark for getting an expression for x or y and correctly differentiating

1 mark for the correct use of chain rule.

1 mark for final answer

MATHEMATICS Extension 1 : Question...7....

Suggested Solutions	Marks	Marker's Comments
a) i) $\cos x + \sqrt{3} \sin x = r \cos(x - \alpha)$ $= r \cos x \cos \alpha + r \sin x \sin \alpha$ $r \cos \alpha = 1$ $r \sin \alpha = \sqrt{3}$ $r^2 \cos^2 \alpha + r^2 \sin^2 \alpha = 1 + 3$ $r^2 (\cos^2 \alpha + \sin^2 \alpha) = 4$ $r^2 = 4$ $r = 2$ (since $r > 0$) Method 1: $\frac{r \sin \alpha}{r \cos \alpha} = \frac{\sqrt{3}}{1}$ $\tan \alpha = \sqrt{3}$ $\alpha = \pi/3, (0 < \alpha < \pi/2)$ Method 2: Sub $r=2$ into $r \cos \alpha = 1$ $\cos \alpha = 1/2$ $\alpha = \pi/3 (0 < \alpha < \pi/2)$ Method 3: Sub $r=2$ and $\alpha = \pi/3$ into $r \cos(x - \alpha)$ $2 \cos(x - \pi/3) = 2 \cos x \cos \pi/3 + 2 \sin x \sin \pi/3$ $= 2 \cos x \times 1/2 + 2 \sin x \times \sqrt{3}/2$ $= \cos x + \sqrt{3} \sin x$ $\therefore r=2$ and $\alpha = \pi/3$ satisfies	1 1	showing value of r showing value of α
ii) $\cos x + \sqrt{3} \sin x = 2 \cos(x - \pi/3)$ range: $-2 \leq y \leq 2$ iii) When $x=0$, $y = 2 \cos(-\pi/3)$ $= 2 \times 1/2$ $= 1$ y-intercept is 1	1 1	

MATHEMATICS Extension 1 : Question.....7

Suggested Solutions

Marks

Marker's Comments

iv) when $y=0$.

$$2 \cos(x - \pi/3) = 0$$

$$x - \pi/3 = \pi/2, 3\pi/2$$

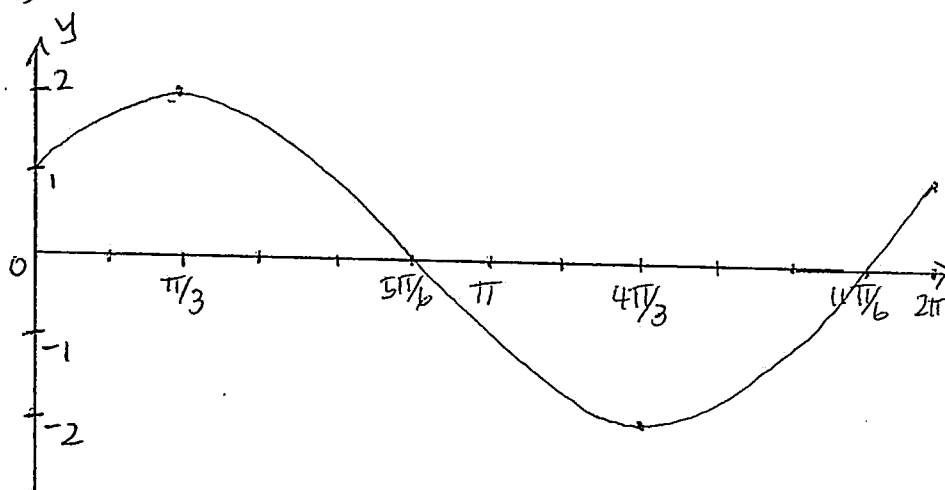
$$x = \pi/2 + \pi/3, 3\pi/2 + \pi/3$$

$$= 5\pi/6, 11\pi/6$$

2

1 for each value of x within given domain

v)



1

3 intercepts and shape

1

max/min points

$$vi) V = \pi \int_0^{5\pi/6} (\cos x + \sqrt{3} \sin x)^2 dx$$

$$= \pi \int_0^{5\pi/6} 4 \cos^2(x - \pi/3) dx$$

$$= 4\pi \int_0^{5\pi/6} \frac{1}{2} (\cos 2(x - \pi/3) + 1) dx$$

$$= 2\pi \int_0^{5\pi/6} (\cos(2x - 2\pi/3) + 1) dx$$

$$= 2\pi \left[\frac{1}{2} \sin(2x - 2\pi/3) + x \right]_0^{5\pi/6}$$

$$= \pi [\sin(5\pi/3 - 2\pi/3) + 5\pi/3 - \sin(-2\pi/3) - 0]$$

$$= \pi [\sin \pi + 5\pi/3 + \sqrt{3}/2]$$

$$= \pi [5\pi/3 + \sqrt{3}/2] \text{ cubic units}$$

}

1

Correct expression for volume

1

change to $\cos 2\theta$

1

Correct substitution and evaluation

a) $D: x > 1$ $\mid$
 $R: y \in \mathbb{R}$ $\mid$

b) $\frac{x+4-x-1}{(x+1)(x+4)} = \frac{3}{(x+1)(x+4)} \quad \therefore a=3 \mid$

c) $\int_0^1 \frac{dx}{(x+1)(x+4)} = \frac{1}{3} \int \left(\frac{1}{x+1} - \frac{1}{x+4} \right) dx$
 $= \frac{1}{3} \left[\ln \left(\frac{x+1}{x+4} \right) \right]_0^1 \mid$
 $= \frac{1}{3} \left(\ln \frac{2}{5} - \ln \frac{1}{4} \right) \mid$
 $\approx \frac{1}{3} \ln \left(\frac{2 \times 4}{5} \right) = \frac{1}{3} \ln \left(\frac{8}{5} \right)$
 $\approx \frac{1}{3} (\ln 2 - \ln 5 + \ln 4)$

c) $\int_0^{\frac{\pi}{6}} 2 \cos \theta \cdot 2 \cos \theta d\theta$
 $= 4 \int_0^{\frac{\pi}{6}} \cos^2 \theta d\theta \mid$
 $= \frac{4}{2} \int_0^{\frac{\pi}{6}} (1 + \cos 2\theta) d\theta \mid$
 $= 2 \left(\theta + \frac{\sin 2\theta}{2} \right) \Big|_0^{\frac{\pi}{6}}$
 $= 2 \left(\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right)$
 $= \frac{\pi}{3} + \frac{\sqrt{3}}{2}$

$x = 2 \sin \theta$
 $dx = 2 \cos \theta d\theta$
 $x=0, \theta=0$
 $x=1, \theta=\frac{\pi}{6}$

1 for correct integration
 double angle and
 substitute values of θ

$$d) \quad y = \tan x^2$$

$$y' = 2x \sec^2(x^2) \quad |$$

$$\int_{-a}^a x \sec^2(x^2) dx$$

$$= \left[\frac{1}{2} \tan(x^2) \right]_{-a}^a \quad |$$

$$= \frac{1}{2} \tan a^2 - \frac{1}{2} \tan(-a)^2$$

$$= \underline{0} \quad |$$

or

$f(x) = x \sec^2 x^2$ is an odd function since

$$f(-x) = -x \sec^2(-x)^2$$

$$f(-x) = -x \sec^2 x^2$$

$$\therefore f(x) = -f(-x)$$

Suggested Solutions

Marks

Marker's Comments

a) $T_4 = a + 3d = 5$

$\therefore a = 5 - 3d$

Let $S = T_1^2 + T_2^2 + T_3^2$

$= a^2 + (a+d)^2 + (a+2d)^2$

$= 3a^2 + 5d^2 + 6ad$

$= 3(5-3d)^2 + 5d^2 + 6d(5-3d)$

$= 75 - 60d + 14d^2$

$\frac{dS}{dd} = -60 + 28d$

$= 0$ at Turning Point

$\therefore d = \frac{15}{7}$ " " "

$\frac{d^2S}{dd^2} = 28$

$> 0 \therefore$ Concave up

$\therefore S$ is a minimum when
 $d = \frac{15}{7}$

Rather too many people used calculus before eliminating a (or d). Such calculus is invalid as a and d vary together to satisfy $T_4 = 5$.

Such people did not usually qualify for any marks.

A few people substituted for d rather than a . It should have worked but the numbers were uglier.

Suggested Solutions	Marks	Marker's Comments
<u>To Prove</u> : $\frac{1}{3} \times \frac{1}{1} + \frac{1}{5} \times \frac{1}{3} + \dots + \frac{1}{(2n+1)} \times \frac{1}{(2n-1)} = \frac{n}{2n+1}$ <u>Step 1</u> : Check for $n=1$. $\text{LHS} = \frac{1}{3} \times \frac{1}{1} \quad \text{RHS} = \frac{1}{2+1}$ $= \frac{1}{3} \quad = \frac{1}{3}$ LHS = RHS so <u>true for $n=1$.</u> <u>Step 2</u> : Assume true for $k \in \mathbb{Z}^+$ i.e. $\frac{1}{3} \times \frac{1}{1} + \frac{1}{5} \times \frac{1}{3} + \dots + \frac{1}{(2k+1)} \times \frac{1}{(2k-1)} = \frac{k}{2k+1}$ It is required to prove for $(k+1)$. i.e. $\frac{1}{3} \times \frac{1}{1} + \dots + \frac{1}{(2k+1)} \times \frac{1}{(2k-1)} + \frac{1}{(2k+3)} \times \frac{1}{(2k+1)} = \frac{k+1}{2k+3}$ LHS = $\frac{k}{2k+1} + \frac{1}{(2k+3)(2k+1)}$ (by Assumption above) $= \frac{k(2k+3) + 1}{(2k+3)(2k+1)}$ $= \frac{2k^2 + 3k + 1}{(2k+3)(2k+1)}$ $= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} = \frac{(k+1)}{(2k+3)} \text{ as req'd}$ <u>Step 3</u> : Using steps 1 and 2 and invoking the Principle of Mathematical Induction, the proof is complete for $n \geq 1$.	1 1 1 1	One for use of the word <u>Assumption</u> or something equivalent was required. Some very short cuts taken with the last step.

Suggested Solutions

Marks

Marker's Comments

c) $y = \frac{24}{(x+6)(x-4)}$, $\frac{dy}{dx} = \frac{-24(2x+2)}{(x^2 - 2x - 24)^2}$

1

Asymptotes (3) must be drawn and their equations must be found somewhere, preferably on the diagram.

Vertical asymptotes $x = -6$, $x = 4$

Horizontal asymptote $y = 0$

No x intercept

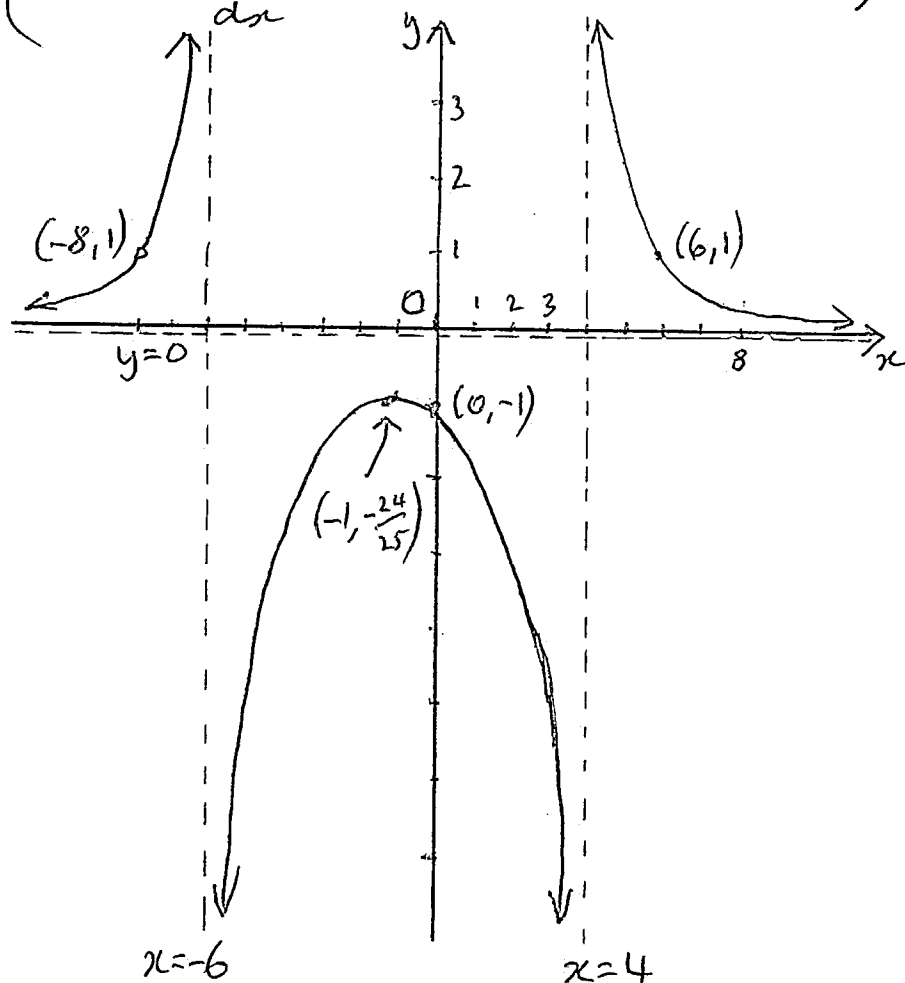
y intercept at $(0, -1)$

Turning point at $(-1, -\frac{24}{25})$

(where $\frac{dy}{dx} = 0$. It is a maximum)

1

y intercept must be marked on diagram, preferably in coordinate form $(0, -1)$



1

Turning point similarly $(-1, -\frac{24}{25})$

1

Three parts of the shape basically correct. The max point must be above the intercept.

(If there is no max it would be impossible to get more than 2 marks)

Suggested Solutions

Marks Awarded

Marker's Comments

ai) Method 1
 $x^2 = 4Ay$
 $y = \frac{x^2}{4A}$

$$\frac{dy}{dx} = \frac{2x}{4A} = \frac{x}{2A}$$

when $x = 2Ap$

$$\frac{dy}{dx} = \frac{2Ap}{2A} = p$$

①

Method 2

$$x = 2Ap \quad y = Ap^2$$

$$\frac{dx}{dp} = 2A \quad \frac{dy}{dp} = 2Ap$$

$$\frac{dy}{dx} = \frac{dy}{dp} \times \frac{dp}{dx}$$

$$= 2Ap \times \frac{1}{2A} = p$$

①

Equation of tangent

$$y - y_1 = m(x - x_1)$$

$$y - Ap^2 = p(x - 2Ap)$$

$$y - Ap^2 = px - 2A^2p$$

$$y - px + A^2p = 0$$

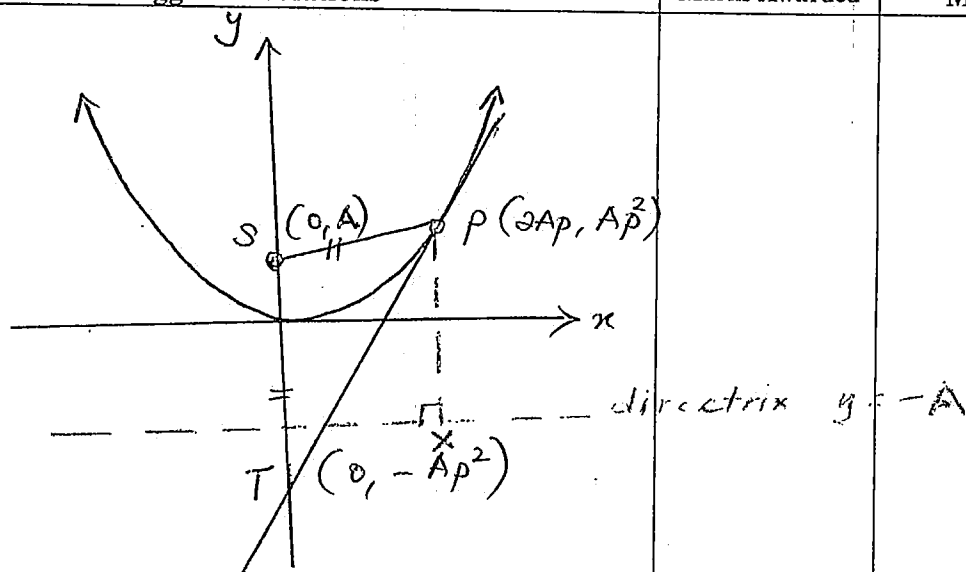
ie $0 = px - y - Ap^2$
 (shown as required)

Suggested Solutions

Marks Awarded

Marker's Comments

ii)



$$\text{RTP } SP = ST$$

Method 1

from part (i)

$$px - y - Ap^2 = 0 \quad \text{--- i}$$

 T is on y axis $\therefore x = 0$

$$\text{Thus } y = -Ap^2$$

 so coord T is $(0, -Ap^2)$

$$\begin{aligned} \text{d } ST &= A + Ap^2 \quad (\text{vertical length}) \\ &= A(1 + p^2) \quad \text{--- (i)} \end{aligned}$$

$$\begin{aligned} \text{d } SP &= \sqrt{(Ap^2 - A)^2 + (2Ap - 0)^2} \\ &= \sqrt{A^2p^4 - 2A^2p^2 + A^2 + 4A^2p^2} \\ &= \sqrt{A^2p^4 + 2A^2p^2 + A^2} \\ &= \sqrt{A^2(p^4 + 2p^2 + 1)} \\ &= \sqrt{A^2(1 + p^2)^2} > 0 \\ &= A(1 + p^2) \quad \text{--- (i)} \end{aligned}$$

$$\therefore SP = ST$$

Suggested Solutions

Marks Awarded

Marker's Comments

Method 2

$$\begin{aligned} d \text{ SP} &= \sqrt{(Ap^2 - A)^2 + (2Ap - 0)^2} \\ &= \sqrt{A^2p^4 - 2A^2p^2 + A^2 + 4A^2p^2} \\ &= \sqrt{A^2p^4 + 2A^2p^2 + A^2} \end{aligned}$$

$$\begin{aligned} d \text{ ST} &= \sqrt{(-Ap^2 - A)^2 + (0 - 0)^2} \\ &= \sqrt{A^2p^4 + 2A^2p^2 + A^2} \\ &= A(1+p^2) \end{aligned}$$

Some stopped here

* must have both eqns looking the same coz its RTT

Method 3

$SP = Px$ (by definition of parabola)
(refer diagram)

$$\begin{aligned} \therefore SP &= A + Ap^2 \\ &= A(1+p^2) \end{aligned}$$

$$\begin{aligned} ST &= A + Ap^2 \\ &= A(1+p^2) \end{aligned}$$

$$\therefore SP = ST$$

1 mark to show coord of T

1 mark for correct SP

1 mark for correct ST

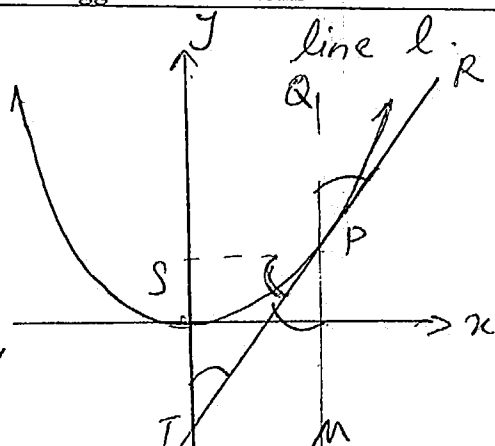
* and both SP and ST must be the same by correct working out

Suggested Solutions

Marks Awarded

Marker's Comments

iii)



Method 1

Let line l be parallel to y axis

from pt ii) $ST = SP$

$\therefore \angle SPT = \angle STP$ (angles opposite equal sides in a triangle are equal)

$\angle STP = \angle TPM$ (alt angle $ST \parallel PM$)

$\therefore \angle SPT = \angle TPM$

OR .

$\angle STP = \angle QPR$ (corresponding angles $ST \parallel PM$)

$\therefore \angle SPT = \angle QPR$

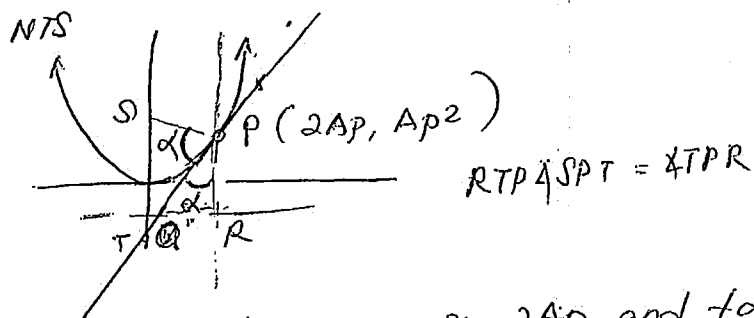
①

Suggested Solutions

Marks Awarded

Marker's Comments

Method 2


 Angle between $x = 2Ap$ and tangent

$$\tan \alpha = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$m_{SP} = \frac{Ap^2 - A}{2Ap - 0} \quad m_{PT} = \frac{Ap^2 - (-Ap^2)}{2Ap - 0}$$

$$= \frac{p^2 - 1}{2p} \quad = p$$

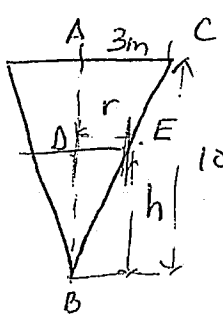
$$\tan \alpha = \frac{\frac{p^2 - 1}{2p} - p}{1 + \left(\frac{p^2 - 1}{2p} \times p\right)}$$

$$= \frac{1}{p}$$

$$\angle SPT \therefore \alpha = \tan^{-1} \frac{1}{p} \quad \text{--- (1)}$$

$$\angle TPR = \pi - \angle PQR - \frac{\pi}{2} \quad (\text{angle sum of } \triangle PQR)$$

$$= \frac{\pi}{2} - \angle PQR$$

Suggested Solutions	Marks Awarded	Marker's Comments
$\tan \angle TPR = \tan \left(\frac{\pi}{3} - \angle PQR \right)$ $= \cot \angle PQR$ $\tan \angle PQR = m_{TP} = p$ $\cot \angle PQR = \frac{1}{p}$ $\therefore \tan \angle TPR = \frac{1}{p}$ $\therefore \tan \angle SPT = \tan \angle TPR$ $\angle SPT = \angle TPR \quad \text{both angles equal.}$ b)  $\frac{dV}{dh} = 2\pi r^2 \frac{dr}{dh} = 2\pi r^2 \cdot \frac{r}{h} = \frac{2\pi r^3}{h}$ Method 1 i) Show $r = \frac{3h}{10}$. Must! state $(\triangle DBE \parallel \triangle ABC)$ or (state using similar triangles) to get the mark. $\frac{h}{r} = \frac{10}{3}$ $\therefore r = \frac{3h}{10}$ Method 2 Let $\angle DBE = \theta$ $\tan \theta = \frac{r}{h}$ $\tan \theta = \frac{3}{10} \quad \therefore r = \frac{3h}{10}$		* If used rhombus needed to say 1 pair of adjacent sides equal and 1 pair of opposite sides are equal & parallel ①

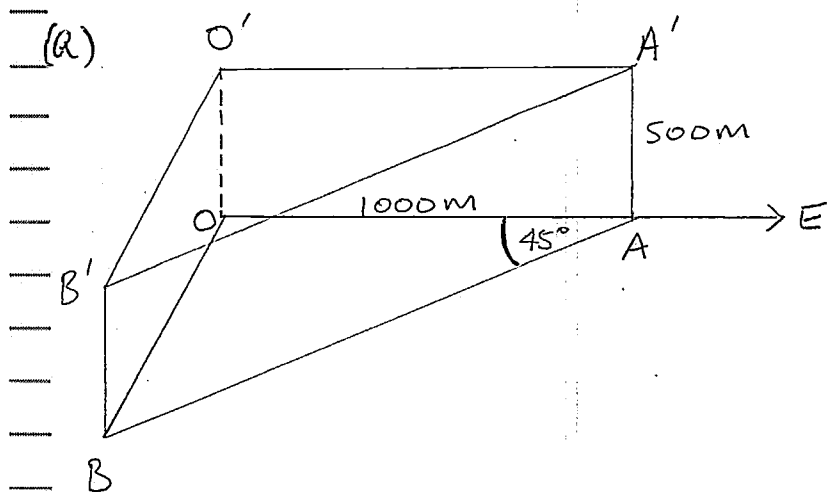
Suggested Solutions	Marks Awarded	Marker's Comments
ii) $V = \frac{1}{3} \pi r^2 h$ $= \frac{1}{3} \pi \left(\frac{3h}{10} \right)^2 h$ $= \frac{\pi h}{3} \left(\frac{9h^2}{100} \right)$ $= \frac{3\pi h^3}{100}$ ——— (1) $\frac{dV}{dh} = \frac{3 \times 3\pi h^2}{100} = \frac{9\pi h^2}{100}$ $\frac{dh}{dt} = ? \quad \frac{dV}{dt} = 2$ $\therefore \frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$ $2 = \frac{9\pi h^2}{100} \times \frac{dh}{dt}$ $\frac{dh}{dt} = \frac{100 \times 2}{9\pi h^2}$ $= \frac{200}{9\pi h^2}$ sub $h=8$ $= \frac{200}{9 \times \pi \times 8 \times 8}$ $= \frac{25}{72\pi}$ m/min ——— (1)	final answer ————— (1) ————— (1)	* You cannot differentiate with 2 variables. - if you have done so, max mark is 1/3 * if there arent any subsequent error - you will lose a mark if your units were incorrect. - it has to be below the vinculum.

MATHEMATICS Extension 1 : Question...!!

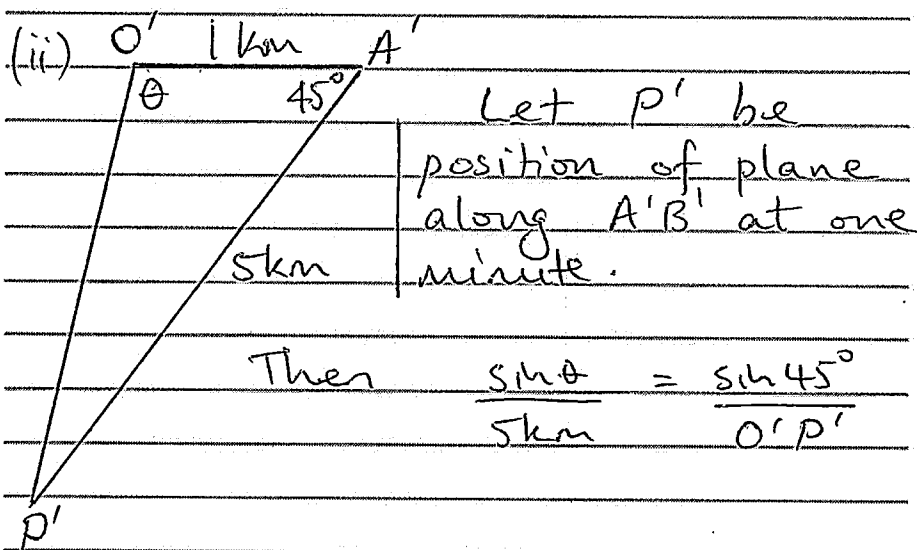
Suggested Solutions

Marks

Marker's Comments



$$\begin{aligned}
 \text{(i)} \quad \frac{300 \text{ km}}{h} &= \frac{300 \text{ km}}{60 \text{ min}} \\
 &= 5 \text{ km/min} \\
 &= 5000 \text{ m/min}
 \end{aligned}$$



Main error: incorrect order of magnitude.

Many had difficulty converting 3D problem into 2D components.

MATHEMATICS Extension 1 : Question...!!...

Suggested Solutions

Marks

Marker's Comments

Now,

$$(O'P')^2 = (1\text{km})^2 + (5\text{km})^2$$

$$- 2(1\text{km})(5\text{km}) \cos 45^\circ$$

$$= 26 - 10/\sqrt{2} \text{ km}^2$$

$$= 26 - 5\sqrt{2} \text{ km}^2$$

So,

$$O'P' = \sqrt{26 - 5\sqrt{2}} \text{ km}$$

Hence

$$\sin \theta = \sin 45^\circ \cdot \frac{5\text{km}}{\sqrt{26 - 5\sqrt{2}}}$$

$$\Rightarrow \theta \doteq 54^\circ 21'$$

OR

$$180^\circ - 54^\circ 21' \\ = 125^\circ 39'$$

Since $125^\circ 39' + 45^\circ < 180^\circ$,

$125^\circ 39'$ is a possibility.

*Main error:
mixing units/
dropping units
leading to
incorrect
quantity.

*Other: incorrect
use of cosine
rule.

*Leave answers
in exact form
as long as possible
- marker can follow;
- avoid compounding
errors.

*Most did not

consider the
possible comp-
lementary case.
Must check.
Also, should
justify why one
of two angles
is chosen.

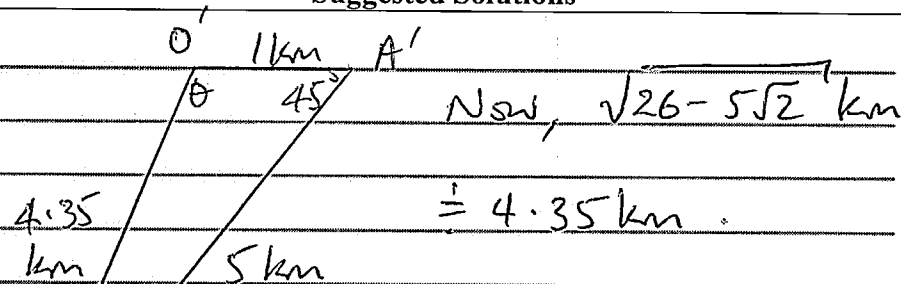


MATHEMATICS Extension 1 : Question...!!

Suggested Solutions

Marks

Marker's Comments



The triangle indicates that θ should be the largest angle since θ is opposite the largest side.

If $\theta = 54^\circ 21'$, then

$$\alpha = 180^\circ - 45^\circ - \theta$$

$$= 80^\circ 39'$$

meaning $\alpha > \theta$ //

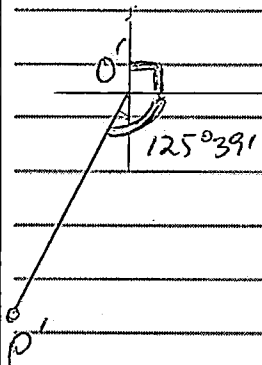
$$\therefore \theta = 125^\circ 39'$$

Therefore, the bearing of P' from O' is

$$90^\circ + 125^\circ 39'$$

$$= 215^\circ 39'$$

//



* When $\triangle$ has three sides given/determined, could use cosine rule & avoid 'trouble' with sine rule.

* Good idea to include diagrams in any geometrical proof - helps your case/makes your argument clearer.

* Some failed to recognise that 90° needed to be added for bearing

MATHEMATICS Extension 1 : Question.....

Suggested Solutions

Marks

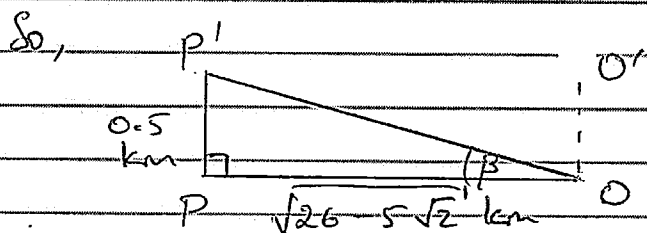
Marker's Comments

(iii) P' lies at a distance

$$\sqrt{26 - 5\sqrt{2}} \text{ km}$$

from O' & it is

0.5 km in altitude.



$\angle POP' = \beta$ will be angle of elevation.

$$\text{Then } \tan \beta = \frac{1/2}{\sqrt{26 - 5\sqrt{2}} \text{ km}}$$

$$\Rightarrow \beta \doteq 6^\circ 33'$$

* Main issue:
people mixing
units. E.g.

$$\tan \beta = \frac{500}{4.35}$$

not recognising
that '500'
is 500 m &
'4.35' is
4.35 km.

Either

- make quantities
uniform (same
units)

or

- Keep units
in the
expression.

MATHEMATICS Extension 1 : Question...

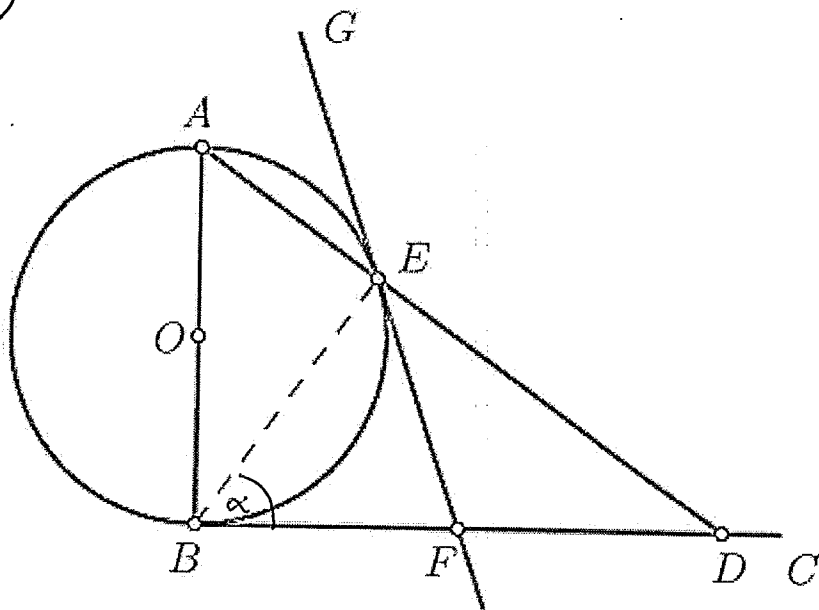
Suggested Solutions

Marks

Marker's Comments

(b)

(i)



(ii) (1) Let $\angle EBF = \alpha$.

(2) $\angle ABF = \pi/2$ (Diameter of circle is perpendicular to tangent at point of contact).

(3) $\angle ABF = \angle ABE + \angle EBF$ (adjacent angle sum)

i.e. $\frac{\pi}{2} = \angle ABE + \alpha$ (by (1) & (2))

so, $\angle ABE = \frac{\pi}{2} - \alpha$

Mark awarded for appropriate start.

→

MATHEMATICS Extension 1 : Question...!!...

Suggested Solutions

Marks

Marker's Comments

(4) $\angle GEA = \angle ABE$ (angle between tangent & chord equals the angle at the circumference in the alternate segment).

$$\text{So, } \angle GEA = \frac{\pi}{2} - \alpha \quad (\text{by (3)}).$$

(5) $\angle GEA = \angle FED$ (vertically opposite angles).

$$\text{Hence } \angle FED = \frac{\pi}{2} - \alpha \quad (\text{by (4)}).$$

* Every time you make a claim in a geometry proof, you must state why it is true.

Don't need to justify algebraic or arithmetic operations.

MATHEMATICS Extension 1 : Question...!!

Suggested Solutions

Marks

Marker's Comments

(iii) (1) Let $\angle EBF = \alpha$.

(2) BD is tangent (given),

so,

$$\angle ABD = \frac{\pi}{2} \quad (\text{diameter of circle is perpendicular to tangent at point of contact}).$$

(3) Also, $\angle BAD = \angle EBF$

(angle between tangent & chord equals the angle at the circumference in the alternate segment)

$\therefore$ As $\alpha = \angle EBF$ (by (1)),

$$\angle BAD = \alpha$$

$$(4) \angle BAD + \angle ABD + \angle ADB = \pi$$

(angle sum plane triangle)

$$\text{So, } \alpha + \frac{\pi}{2} + \angle ADB = \pi$$

by (3), (2) respectively,

$$\text{i.e. } \angle ABD = \frac{\pi}{2} - \alpha$$

* One mark for appropriate start.

* Don't mix degrees & radians in same expression (unless you attach units).

MATHEMATICS Extension 1 : Question.....11

Suggested Solutions

Marks

Marker's Comments

(5) As $\angle FED = \frac{\pi}{2} - \alpha$

(proven in part (ii))

& $\angle ADB = \frac{\pi}{2} - \alpha$ (by (4)),

$EF = FD$ (equal sides
opposite equal
angles in
triangle)

(6) Also, both BF , EF are
tangents (given), so,

$BF = EF$ (intercepts on
tangents drawn
from a common
point to a circle
are equal).

(7) Hence by (5) & (6),

$FD = EF = BF$

$\therefore BF = FD$

as required. //

* Much of the
script (hand-
writing) was
ILLEGIBLE.

Care must
be taken!

* Present ^{can't} your
argument
clearly ;

link relevant
statements
directly.

Don't make
assessor 'fish
around' for
facts to prove
your case

- that's

your job.

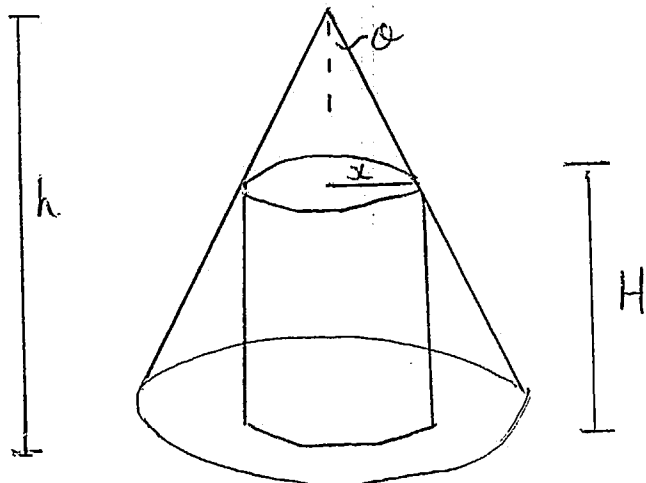
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Suggested Solutions

Marks

Marker's Comments

12. a) i



$$V_{\text{cylinder}} = \pi x^2 H$$

$$\tan \theta = \frac{x}{h-H}$$

$$\therefore \cot \theta = \frac{h-H}{x}$$

$$h-H = x \cot \theta$$

$$H = h - x \cot \theta$$

$$\begin{aligned} \therefore V_{\text{cylinder}} &= \pi x^2 (h - x \cot \theta) \\ &= \pi (x^2 h - x^3 \cot \theta) \end{aligned}$$

Alternate Method

Let the radius of cone be R

$$\frac{h-H}{h} = \frac{x}{R} \quad (\text{corresponding sides of similar triangles})$$

$$\text{but } R = h \tan \theta$$

(1)

(1)

(1)

Suggested Solutions

Marks

Marker's Comments

$$\therefore \frac{h-H}{h} = \frac{x}{h \tan \theta}$$

$$\therefore (h-H) \tan \theta = x$$

$$h-H = x \cot \theta$$

$$H = h - x \cot \theta$$

$$\therefore V_{\text{cylinder}} = \pi x^2 (h - x \cot \theta)$$

$$V_{\text{cylinder}} = \pi (x^2 h - x^3 \cot \theta)$$

$$\text{ii} \quad V = \pi (x^2 h - x^3 \cot \theta)$$

$$\frac{dV}{dx} = \pi (2xh - 3x^2 \cot \theta)$$

$$\frac{dV}{dx} = 0 \Rightarrow 2xh - 3x^2 \cot \theta = 0$$

$$x(2h - 3x \cot \theta) = 0$$

$$\therefore x = 0 \text{ or } x = \frac{2h \tan \theta}{3}$$

①

But x is a length and greater than 0.

$$\therefore x = \frac{2h \tan \theta}{3} \text{ only.}$$

$$\frac{d^2V}{dx^2} = \pi (2h - 6x \cot \theta)$$

$$\text{When } x = \frac{2h \tan \theta}{3},$$

$$\frac{d^2V}{dx^2} = \pi \left(2h - \frac{6 \cot \theta \times 2h \tan \theta}{4} \right)$$

$$= -2\pi h$$

$$< 0 \quad \therefore \text{maximum at } x = \frac{2h \tan \theta}{3}$$

①

Suggested Solutions

Marks

Marker's Comments

When $x = \frac{2h \tan \theta}{3}$

$$H = h - \left(\frac{2h \tan \theta}{3} \right) \cot \theta$$

$$= h - \frac{2h}{3}$$

$$H = \frac{h}{3} \text{ gives a maximum}$$

①

b) i) $\cos(a+b) = \cos a \cos b - \sin a \sin b,$

When $a=b=q,$

$$\cos(2q) = \cos q \cos q - \sin q \sin q$$

$$= \cos^2 q - \sin^2 q$$

$$= (1 - \sin^2 q) - \sin^2 q$$

$$= 1 - 2\sin^2 q$$

①

ii) LHS = $\frac{\cos y - \cos(y+2q)}{2\sin q}$

$$= \frac{\cos y - (\cos y \cos 2q - \sin y \sin 2q)}{2\sin q}$$

①

$$= \frac{\cos y (1 - \cos 2q) + \sin y \sin 2q}{2\sin q}$$

$$= \frac{\cos y \frac{2\sin^2 q}{2} + \sin y \frac{2\sin q \cos q}{2\sin q}}$$

$$= \cos y \sin q + \sin y \cos q$$

$$= \sin(y+q)$$

①

Suggested Solutions	Marks	Marker's Comments
iii) Test for $n=1$, $\text{LHS} = \sin q$ $\text{RHS} = \frac{1 - \cos 2q}{2 \sin q}$ $= \frac{1 - [1 - 2 \sin^2 q]}{2 \sin q}$ $= \frac{2 \sin^2 q}{2 \sin q}$ $= \sin q$ $= \text{LHS}$ $\therefore$ The result is true for $n=1$		①
Assume the result is true for $n=k$, $k \in \mathbb{Z}^+$ $\therefore \sin q + \sin 3q + \dots + \sin (2k-1)q = \frac{1 - \cos 2kq}{2 \sin q}$ <u>Method 1</u> Test for $n=k+1$, $\therefore \text{LHS} = \sin q + \sin 3q + \dots + \sin (2k-1)q + \sin (2k+1)q$ $= \frac{1 - \cos 2kq}{2 \sin q} + \sin (2kq + q) \text{ (By assumption)}$ $= \frac{1 - \cos 2kq}{2 \sin q} + \frac{\cos(2kq) - \cos(2kq + 2q)}{2 \sin q} \text{ (from part ii where } y=2kq \text{)}$ $= \frac{1 - \cos 2(k+1)q}{2 \sin q}$ $\therefore$ If the result is true for k, it is true for $k+1$, $\therefore$ The result is true by Mathematical Induction.		①

Suggested Solutions	Marks	Marker's Comments
iii) Alternate Method Test for $n=k+1$ LHS = $\sin q + \sin 3q + \dots + \sin(2k-1)q + \sin(2k+1)q$ $= \frac{1 - \cos 2kq}{2 \sin q} + \sin(2kq + q) \text{ (By assumption)}$ $= \frac{1 - \cos 2kq}{2 \sin q} + (\sin 2kq \cos q + \cos 2kq \sin q)$ $= \frac{1 - \cos 2kq + 2 \sin q \cos q \sin 2kq + 2 \sin^2 q \cos 2kq}{2 \sin q}$ $= \frac{1 - \cos 2kq + \sin 2q \sin 2kq + (1 - \cos 2q) \cos 2kq}{2 \sin q}$ $= \frac{1 - \cos 2kq + \sin 2q \sin 2kq + \cos 2kq - \cos 2q \cos 2kq}{2 \sin q}$ $= \frac{1 - (\cos 2q \cos 2kq - \sin 2q \sin 2kq)}{2 \sin q}$ $= \frac{1 - \cos(2kq + 2q)}{2 \sin q}$ $= \frac{1 - \cos 2(k+1)q}{2 \sin q}$ $\therefore$ If the result is true for k, it is true for $k+1$. $\therefore$ The result is true by Mathematical Induction	(1) (1)	